

DS120C: Programming for Data Science

Group Project Report

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Analysis of Factors Affecting Student Exam Performance

The project is meant to examine the effect of some factors on students academic performance with the help of a real-world dataset called StudentPerformanceFactors. It analyzes how aspects such as attendance, sleep hours, study hours, internet access etc. affect students' exam grades.

The project was implemented through Python and follows an OOP. Data analysis and preprocessing was done with the help of NumPy and Pandas libraries, and the data visualization was implemented through Matplotlib.

Dataset Description

The dataset includes information of students and factors which influence their academic performance. It consists of both numerical and categorical data, for which it is applicable for statistical analysis and visualization.

Columns in the dataset contain data of study hours, attendance, sleep hours, motivation level, previous courses, internet access, physical activity, tutoring sessions and exam scores.

The dataset was loaded from a CSV file and handled through separate Python modules.

dataset_handler.py

The file above is responsible for loading and preprocessing the dataset. The following preprocessing steps were implemented:

- loading the dataset using pandas
- Checking basic dataset information
- Creating artificial missing values for preprocessing demonstration
- Handling missing values using NumPy median and mode
- Removing duplicate rows
- Creating additional grouped columns for analysis

data_analyzer.py

This file includes all functions for analysis and visualization such as:

- Statistical summary using describe()
- Grouping and aggregation analysis
- Correlation analysis
- Multiple visualizations using Matplotlib

main.py

This file is responsible for connecting all classes and runs the complete project.

1.Data Preprocessing

During the project, several missing values were added intentionally in order to demonstrate preprocessing techniques. Numerical missing values were filled using the median value calculations with NumPy, while categorical values were filled using the mode. Duplicate rows were also extracted to improve data quality. Some additional columns such as score levels and study hour groups were attached to make the analysis more meaningful.

2.Analysis and Results

Several analyses were implemented regarding the fluctuations of students' exam grades.

3.Statistical Summary

A statistical summary provides information such as mean, minimum, maximum, and standard deviation for numerical columns.

4.Grouping and Aggregation

The project examined average exam grades based on:

- Gender
- Motivation level
- Internet access

The results revealed that students with higher motivation level and internet access show higher academic performance and therefore have higher grades.

5.Correlation Analysis

Correlation analysis determined which factors were most related to exam scores.

The analysis revealed:

- Attendance and hours studied had a positive influence on exam scores.
- Sleep hours had very weak correlation with performance.
- Previous scores also had a noticeable influence on final exam results.

6. Visualizations

Various types of visualizations were implemented to better depict the data.

The project includes:

- Histogram of exam score distribution
- Bar charts comparing average scores
- Scatter plot of study hours versus exam score
- Line chart of attendance ranges and average scores
- Correlation heatmap

For further graph customizations colors, grids and dashed median lines were used to make them more readable.

Summary

The overall purpose of the project was to demonstrate how data analysis techniques can be used to examine educational data and identify aspects that influence academic performance.

Moreover, the project demonstrates the practical use of object-orientated programming, modular code organization, and Python libraries such as NumPy, Pandas and Matplotlib.

Through preprocessing, statistical analysis, correlation analysis grouping and visualization, the project depicts the relationship between academic success and real-life factors.

